



Taking into account the change of support when fusing heterogeneous spatial data for interpolation

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Abstract. *This study investigates the integration of remotely sensed covariates, NDVI from Sentinel-2 and Google's Satellite Embedding imagery, into a multivariate geostatistical framework for specialty coffee yield mapping in southeastern Brazil. Coffee yield data were sparse and influenced by biennial alternation, exhibiting outliers and non-normal distributions. Block cokriging (BCOK) was then used to fuse yield observations with satellite-derived covariates under a common spatial support. Variographic analysis showed that dual-covariate BCOK improved spatial structure, extending ranges and reducing nugget effects relative to ordinary kriging. By aligning heterogeneous datasets through support regularization, BCOK produced smoother interpolated surfaces, minimized uncertainty, and enabled more robust yield predictions.*

1. Introduction

Coffee yield depends on numerous biophysical and environmental factors (1). A further challenge in production is the biennial yield pattern, where high and low yields alternate across different areas of the same field (2). Precision Agriculture (PA), as defined by the International Society of Precision Agriculture, is a management strategy that integrates spatial, temporal, and individual agronomic data to optimize inputs and improve both economic and environmental outcomes (3). Because soil properties and crop performance vary spatially, uniform management often proves inefficient (4). Spatial modeling methods, particularly geostatistics, allow farmers to account for this variability and guide site-specific management (2, 5, 6). Critically, recognizing the spatial support of data transforms heterogeneity from a limitation into an opportunity for improved accuracy (7).

Geostatistics provides tools to describe and predict spatially distributed variables. In agriculture, these methods estimate soil and crop attributes at unsampled locations using limited observations (8). Yet small sample sizes and irregular sampling grids reduce reliability, particularly in the presence of outliers (9). Multivariate geostatistics, such as

cokriging, address these limitations by incorporating auxiliary variables that are more densely sampled or remotely sensed (10). High-volume auxiliary data sources—such as multispectral vegetation indices and satellite imagery—can further strengthen spatial models of agricultural variables (11).

At larger scales, monitoring increasingly relies on the fusion of heterogeneous, multi-temporal satellite records (12). Traditional approaches engineer per-sensor spectral features or indices and fit task-specific models, often constrained by limited labels. Recent advances, such as *Satellite Embedding V1*, replace this ad hoc featurization with compact, learned representations at 10 m resolution. Each pixel is represented by a 64-component annual vector summarizing multi-source Earth Observation (EO) time series and contextual information in a unit-sphere embedding space (13). These embeddings serve as reusable foundations for downstream tasks, enabling low-shot learning and efficient similarity computations.

The asynchronous nature of coffee yield, the influence of outliers, and sparse sampling complicate efforts to model spatial dependence (7). We hypothesize that integrating remotely sensed data as covariates in multivariate geostatistical models improves the interpolation of specialty coffee yield. The objective of this study is therefore to evaluate block cokriging (BCOK) with satellite-derived vegetation indices and Satellite Embedding data as auxiliary variables to support yield mapping in a coffee plot characterized by spatial heterogeneity and non-uniform yield distributions.

2. Material and Methods

2.1. Experimental site and settings

The experimental site is located in Paraguaçu, southern Minas Gerais, Brazil (21° 39' 13" S and 45° 48' 07" W) along with the georeferenced soil sampling points collected in May 2021. The area spans 4 ha and is cultivated with *Coffea arabica* L. (cv. Catucaí Amarelo), planted in 2012 at a density of 2000 plants ha⁻¹, with 2.5 m between rows and 1 m between plants, at an altitude of 894.3 m. A total of 40 georeferenced sampling points were defined using a GPS (mean error: 2–5 m), and yield data were collected from subsamples of paired trees at each point. The soil is classified as eutrophic Argisols, which are fertile with favorable physical conditions and moderate slopes, though they may suffer from acidity, high aluminum content, and erosion risks on steeper areas (2). Uniform fertilization was applied directly to the soil in 2021, consisting of 42, 10, and 42 kg·ha⁻¹ of nitrogen (N), phosphorus (P), and potassium (K), respectively (14).

2.2. Multivariate Geostatistics and Block Cokriging

Multivariate geostatistics allows for integrating multiple spatial variables with different supports (15). When variable supports differ significantly in resolution or depth, support regularization is required (12, 16). Here, we standardized all variables to mean zero and unit standard deviation using Gaussian anamorphosis transformation (17), which improves the linearity and optimality of subsequent cokriging predictions (8).

The transformation is defined as:

$$\Phi = \sum \Psi_i H_i(Y) \quad (1)$$

where $H_i(Y)$ are Hermite polynomials and Ψ_i are the corresponding coefficients (10). The inverse transformation is:

$$Y = \Phi^{-1}(Z) \quad (2)$$

which allows reverting the predictions to the original distribution.

Variographic analysis is closely linked to the issue of data support. The spatial resolution of the data used directly influences the shape of the semivariogram:

- Larger supports (e.g., averages per management zone) tend to smooth out variance and shorten the apparent range of spatial dependence;
- Smaller supports (e.g., point measurements) reveal more detail, but increase noise and may highlight small-scale variations.

This relationship is particularly relevant when working with fused data, such as spectral maps with soil data, where support compatibility directly impacts the robustness of variographic analysis and the accuracy of kriging. When the prediction will be made over areas (blocks) rather than points, it is necessary to adjust the transformation to account for the change in support.

Here, the goal is to make the transformation and models compatible with the desired final spatial scale. The spatial structure between all variables (direct and cross) is modeled using LMC. The punctual LMC for different combinations of variables requires regularization over the same block support (here, it is 10 m by 10 m by 0.2 m) by applying block cokriging (BCOK) over the selected block grid (18). This regularization addresses the change of support problem (1, 19, 20).

After block support correction, we fit the Linear Model of Co-Regionalization (LMC) to the direct and cross-variograms, which is a unified model which considers the experimental direct and cross-variograms of all the n variables and then performs weighted least squares (WLS) on the pairs of samples at each lag (20). To perform WLS well, the variables need to be highly correlated. The $n(n+1)/2$ experimental direct and cross-variograms of all the n variables are fitted with a linear combination of the N_S standardized variograms of unit sill, $g^u(\mathbf{u})$. In matrix notation, the LCM follows Equation (3):

$$\Gamma(\mathbf{h}) = \sum_{u=1}^{N_S} \mathbf{B}^u g^u(\mathbf{h}) \quad (3)$$

where $\Gamma(\mathbf{h}) = [\gamma_{ij}(\mathbf{h})]$ is the variogram matrix, and $\mathbf{B}^u = [b_{ij}^u]$ is the positive semi-definite co-regionalization matrix at structure u (20).

Block cokriging (BCOK) was then applied using:

$$Z(B) = \sum_{i=1}^n \lambda_i Z_i + \sum_{j=1}^m \theta_j Y(x_j) + \sum_{j=1}^o \phi_j W(x_j) \quad (4)$$

where Z_i is the observed value, λ_i is the weight, and $Z(B)$ is the predicted value for block B , λ_i are the weights for the primary variable, θ_j are the weights for the first auxiliary variable, and ϕ_j are the weights for the second auxiliary variable.

The weights are obtained by solving the cokriging system based on the variogram of the primary variable ($\gamma_{ZZ}(h)$), the variogram of the secondary variable ($\gamma_{YY}(h)$), and the cross-variogram: ($\gamma_{ZY}(h)$), defined as:

$$\gamma_{ZY}(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)][Y(x_i) - Y(x_i + h)] \quad (5)$$

To enable block predictions, we compute the point-to-block covariance:

$$\overline{C}(B, \mathbf{x}_i) = \int_B \frac{C(v, \mathbf{x}_i)}{|B|} du' \quad (6)$$

where $\overline{C}$ is the block-to-point covariance and C is the point-to-point covariance (21).

BCOK predictions were computed over $10 \text{ m} \times 10 \text{ m} \times 0.2 \text{ m}$ blocks, matching the sample depth.

2.3. Remote Sensing Covariates: NDVI from Sentinel-2 and Satellite Embedding from Google

Normalized Difference Vegetation Index (NDVI, Equation 7) from Sentinel-2 and Satellite Embedding from Google were used as auxiliary variables. NDVI was derived from Sentinel-2 bands:

$$NDVI = \frac{NIR - red}{NIR + red} \quad (7)$$

where NIR and red are reflectances in the near-infrared and red bands, respectively.

NDVI values range from -1 to 1, and areas occupied by denser vegetation tend to present NDVI close to 1 (higher vegetative vigor). NDVI has been applied to detect seasonality effects, phenological stage of vegetation, length of the growing season, peak greenness, and physiological variations of leaves (22). Higher values correspond to healthy vegetation, with a higher density of green biomass (between 0.10 and 1) (23). In exposed soil or less dense vegetation, positive values close to zero are obtained because under this condition there is higher absorption of radiation in the near-infrared band, which explains the low values of NDVI (24).

Satellite Embedding V1 is derived from *AlphaEarth Foundations* (AEF), a geospatial foundation model developed by Google and Google DeepMind. AEF integrates optical (e.g., Sentinel-2, Landsat-8/9), radar (Sentinel-1), and other sources (e.g., topography and climate signals) and learns an *embedding field*—a continuous spatiotemporal representation summarized annually at 10 m. The AlphaEarth preprint reports broad accuracy gains across diverse tasks in sparse-label regimes, along with a compact representation conducive to global-scale processing.(13). Annual global layers for 2017–2024 covering terrestrial land and shallow coastal waters. Each year is a tiled image collection in local UTM projection.(25). Each pixel stores a $\mathbf{z} \in \mathbb{R}^{64}$ with $\|\mathbf{z}\|_2 = 1$. Bands are named A00–A63. Unit-norm embeddings lie on the 63-sphere, enabling cosine-similarity operations and stable clustering (25).

2.4. Model Validation and Uncertainty Assessment

Model performance was assessed using mean error (ME, Equation 8), mean squared error (MSE, Equation 9), kriged reduced mean error (KRME, Equation 10), and kriged reduced mean squared error (KRMSE, Equation 11):

$$ME = \frac{1}{N} \sum_{i=1}^N (Z(x_i) - Z(B, \mathbf{x}_i)) \quad (8)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (Z(x_i) - Z(B, \mathbf{x}_i))^2 \quad (9)$$

$$KRME = \frac{1}{N} \sum_{i=1}^N \frac{Z(x_i) - Z(B, \mathbf{x}_i)}{s} \quad (10)$$

$$KRMSE = \frac{1}{N} \sum_{i=1}^N \left[\frac{Z(x_i) - Z(B, \mathbf{x}_i)}{s} \right]^2 \quad (11)$$

where $Z(x_i)$ is the sampled value at location i , $Z^*(x_i)$ is the predicted value at location i , N is the number of pairs of sampled and predicted values, and s is the standard deviation of the sampled values. The ME and KRME values close to zero indicates a good model performance. MSE indicates good model performance when their values are lower than the variance of the sample values. KRMSE should inside the range $1 \pm (2\sqrt{2})/N$.

2.5. Analysis tools

The geostatistical analyses were performed with Python 3.12 packages SciPy (26) and GSTools (27). We used Google Earth Engine (GEE) to retrieve the ancillary datasets. Final maps and charts were done using Python packages Numpy (28), Geopandas (29), and Matplotlib (30).

3. Results and Discussion

A total of 38 Sentinel-2 images (cloud cover $\leq 25\%$) and 64 Satellite Embedding bands were acquired between June 2020 and May 2021. Their mean pixel values were used to construct NDVI and Satellite Embedding mosaics. Both covariate datasets contained 1817 pixels, a density 45 times greater than the number of coffee yield observations ($N = 40$), underscoring their potential as auxiliary variables. Coffee yield values exhibited non-normality, with high skewness and leptokurtosis, whereas the covariates showed platykurtic distributions (Table 1).

Table 1. Descriptive statistics for coffee yield and covariates

Attribute	N	Min	Mean $\pm$ SD	Max	Kurtosis	Skewness
Coffee yield (kg trees ⁻¹)	40	0.27	3.05 $\pm$ 2.60	16.34	−0.40	0.16
NDVI	1817	0.35	0.53 $\pm$ 0.06	0.64	8.0	−2.02
55th Satellite Embedding band	1817	−10.79	−8.27 $\pm$ 0.60	−6.79	3.95	−0.75

From the embedding set, the 55th band displayed the highest correlation with yield. Pearson coefficients confirmed strong associations with NDVI ($\rho = -0.71$) and the

55th band ($\rho = 0.87$), supporting their inclusion as covariates in block cokriging (BCOK). As most variables showed large departure from normal distribution and were expressed in different units, they were previously transformed to standardized Gaussian variables. Block support correction was then applied to point anamorphosis in order to take into account the change in variance and histogram moving from point to block (change of support). In general, the larger the difference between the original support (v) and the prediction block support (V) ($10 \text{ m} \times 10 \text{ m}$) the smaller the correction coefficient (r) and the larger the reduction in variance (Table 2).

Table 2. Block support correction calculation

Attribute	Punctual variance	Variogram sill	v	r
Coffee yield	1.45	1.94	1.33	0.96
NDVI	1.95	2.44	1.60	0.93
55th Satellite Embedding band	1.22	1.46	1.03	0.94

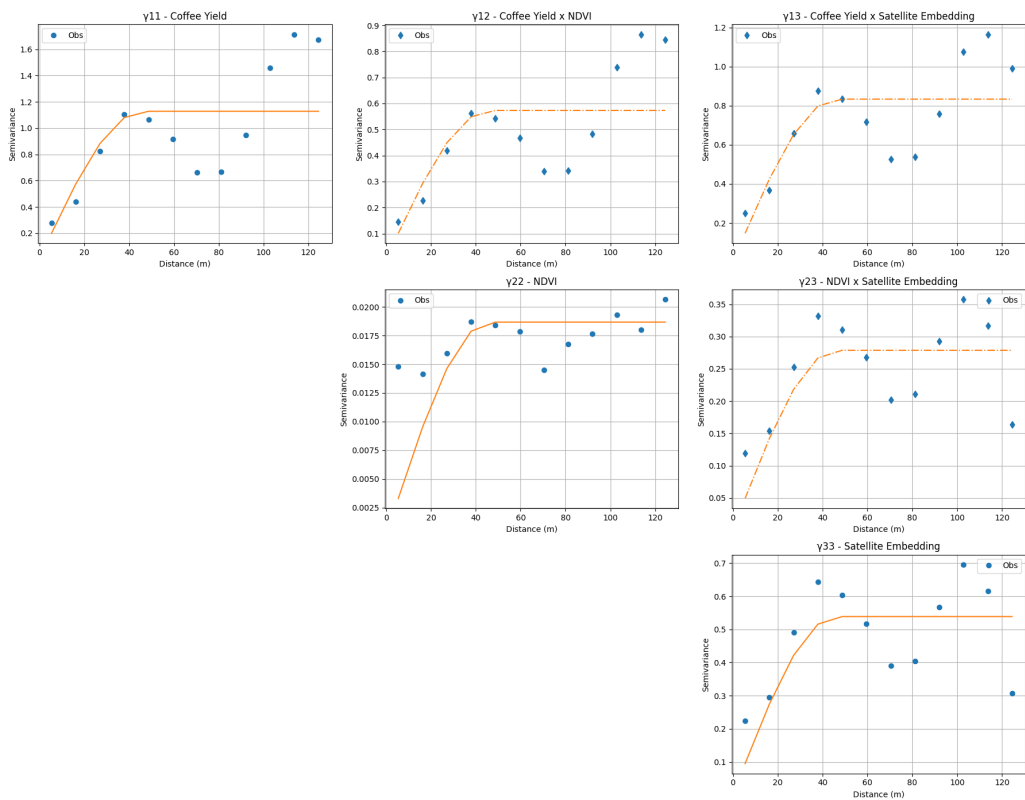


Figure 1. Linear model of coregionalization (LMC) for coffee yield, NDVI, and 55th Satellite Embedding band.

Variographic analysis indicated isotropy, justifying isotropic models. As shown in Figure 1, the fitted linear model of coregionalization captured both direct and cross-variograms, with satisfactory model–data agreement. The inclusion of NDVI and Satellite Embedding extended the range of spatial dependence relative to ordinary kriging (OK) and reduced nugget effects, suggesting stronger spatial continuity and improved representation of field-scale variability (12). When both covariates were used, BCOK achieved

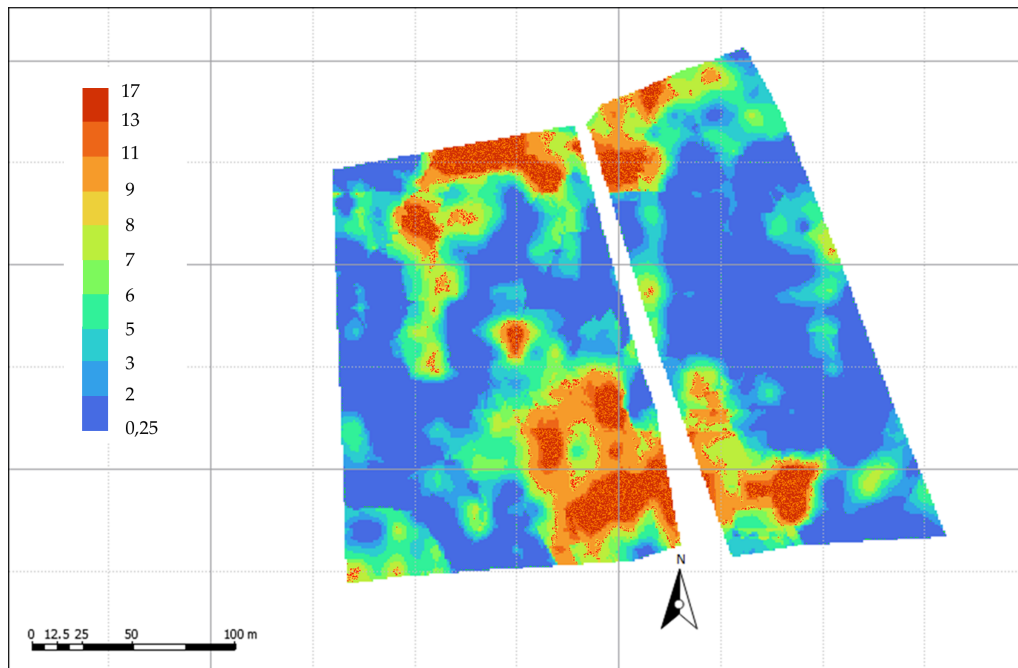


Figure 2. Block cokriging (BCOK) interpolated map of coffee yield (May 2021).

Table 3. Cross-validation of transformed values under ordinary kriging (OK) and block cokriging (BCOK).

Scenario	<i>ME</i>	<i>MSE</i>	<i>KRME</i>	<i>KRMSE</i>
OK (control)	2.33	3.54	3.75	0.75
BCOK with <i>NDVI</i> + Satellite Embedding (band 55)	1.12	1.02	1.11	0.98

the longest range (74.98 m), reflecting enhanced spatial structure. The negative cross-variogram values between yield and *NDVI* were consistent with the negative correlation, while positive values with the embedding matched the strong positive correlation.

Direct relationships between coffee yield and remotely sensed measurements may not be captured by spatial autocorrelation because of the interactions among soil factors (5, 31–33). Only basing the analysis on the comparisons between OK maps and evaluating the Pearson correlation values may be not sufficient to provide insights about coffee yield spatial variability (18). In this manner, multivariate geostatistical analysis can be more useful in disclosing the spatial-scale-dependent correlations between remotely sensed data and yield and consequently provide better coffee yield maps if a relationship between the primary variable and covariates is found. The BCOK-interpolated map (Figure 2) revealed a smoother and more continuous yield distribution than OK. High-yield hotspots were clearly delineated, while low-yield areas appeared less fragmented, indicating reduced estimation variance. This improvement reflects the effective integration of covariates with higher spatial density.

Cross-validation confirmed the superiority of BCOK over OK (Table 3). The dual-covariate model reduced *ME*, *MSE*, and *KRME* to near-zero values and achieved a *KRMSE* (0.98) closest to the ideal. These results demonstrate that integrating satellite-

derived NDVI and embeddings improves interpolation accuracy, even under sparse sampling and biennial yield variability. A similar approach was applied by Silva et al. (18) using NDVI together with Synthetic-aperture radar (SAR) datasets with a little lower accuracy.

The improvements reflect explicit handling of the change of support problem (COSP) (1, 33, 34). Because NDVI and Satellite Embedding data are collected at a finer resolution than yield samples, support regularization was required to ensure comparability. By applying BCOK with a consistent support ($10\text{ m} \times 10\text{ m} \times 0.2\text{ m}$), the model successfully aligned dense remote sensing data with sparse yield observations. This alignment strengthened variogram structures (longer ranges, smaller nuggets) and yielded more reliable predictions.

Overall, these findings highlight the value of COSP-aware modeling in precision agriculture. Properly regularized covariates enabled the fusion of high-resolution Earth Observation datasets with limited field data, producing smoother interpolated surfaces and reducing prediction uncertainty. Beyond methodological rigor, these results emphasize the practical benefit of support regularization: enabling robust use of covariates for site-specific crop management (18). In low-sampled, heterogeneous systems such as coffee, BCOK offers a powerful framework to capture yield variability and inform more precise decision-making.

Conclusions

The integration of multi-source covariates from remote sensing with coffee yield measurements using block cokriging significantly improved spatial predictions in a low-sampled, outlier-prone dataset. The use of NDVI and Satellite Embedding not only enhanced the variogram structure but also reduced interpolation errors and uncertainty. These results demonstrate the potential of multivariate geostatistics, particularly BCOK, for addressing spatial heterogeneity and support mismatch in precision agriculture. By regularizing variables to a common spatial support and leveraging the strength of auxiliary data, the proposed approach provides a robust framework for site-specific yield mapping in specialty coffee production, contributing to more informed and efficient agronomic decision-making.

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